

Learning the theory of Monte Carlo methods

Based on Simon Sirca - Probability for physicists

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1. Introduction

The Monte-Carlo method is a tool used widely in computer simulations in experimental physics. Here, I will study this mathematical and computational technique to enhance my understanding of simulations in High Energy Physics (HEP). The main reference for this learning process is the book *Probability for Physicists* (Širca, 2016).

The Monte-Carlo method considered in this work is the acceptance-rejection method, a technique used to generate random samples from a probability distribution that may be difficult to sample from directly. The method works by generating random samples from a proposal distribution and accepting or rejecting those samples according to a criterion that guarantees that the accepted samples follow the desired target distribution (Nakade, 2025).

1.1. Acceptance-Rejection Method

This will be the method used in the Monte Carlo analysis developed in this repository, since the probability density function (PDF) of the decay spectrum is already known, it can be directly employed as the target distribution in the acceptance rejection sampling method, in our case, the target distribution corresponds to the differential decay rate $\frac{d\Gamma}{dx}$, where x is the normalized electron energy defined as

$$x = \frac{2E_e}{m_\mu}, \quad 0 < x \leq 1,$$

with E_e the electron energy and m_μ the muon mass. Neglecting terms of order m_e^2/m_μ^2 and higher-order radiative corrections $\mathcal{O}(\alpha^2)$, the differential decay rate for free muon decay is given by

$$\frac{d\Gamma^{\text{free}}}{dx} = \frac{G_F^2 m_\mu^5}{192\pi^3} x^2 \left(6 - 4x + \frac{\alpha}{\pi} f(x) \right),$$

where G_F is the Fermi constant and $\alpha = \frac{1}{137.035999173(35)}$ is the fine-structure constant. The function $f(x)$ represents the radiative corrections to the electron energy spectrum and is explicitly written as

$$\begin{aligned} f(x) = & \left[\frac{5}{3x^2} + \frac{16x}{3} + \frac{4}{x} + (12 - 8x) \ln\left(\frac{1}{x} - 1\right) - 8 \right] \ln\left(\frac{m_\mu}{m_e}\right) \\ & + (6 - 4x) \left[2 \text{Li}_2(x) - 2 \ln^2(x) + \ln(x) + \ln(1 - x) \left(3 \ln(x) - \frac{1}{x} - 1 \right) - \frac{\pi^2}{3} - 2 \right] \\ & + \frac{(1 - x)}{3x^2} \left[34x^2 + (5 - 34x^2 + 17x) \ln(x) - 22x \right] \\ & + 6(1 - x) \ln(x), \end{aligned}$$

where $\text{Li}_2(x)$ denotes the dilogarithm (Spence) function. This expression is implemented directly in the Monte Carlo framework and used as the target probability density in the acceptance rejection algorithm for generating electron energy spectra from muon decay (Czarnecki et al., 2014).

Let $p(x)$ be a PDF defined on an interval $[a, b]$, from which we want to sample, suppose that there exists a constant $M > 0$ such that

$$p(x) \leq M, \quad \forall x \in [a, b]. \quad (1)$$

that means, there exists a maximum value of the PDF in the interval, and we can use this maximum value to define a rectangular region in the xy -plane that contains the area under the curve of $p(x)$, then the acceptance-rejection method is based on sampling uniformly from the rectangular region defined by $[a, b] \times [0, M]$, and retaining only those points that lie under the curve $y = p(x)$. According to the fundamental theorem of the method, the marginal distribution of the accepted x values follows the target distribution $p(x)$ (Nakade, 2025).

1.2. Algorithm

The basic acceptance-rejection algorithm can be summarized as follows:

1. **Generate a proposal:** Sample

$$x \sim \mathcal{U}(a, b). \quad (2)$$

2. **Generate a test value:** Sample

$$y \sim \mathcal{U}(0, M). \quad (3)$$

3. **Acceptance criterion:** Accept x if

$$y \leq p(x), \quad (4)$$

otherwise reject the sample and return to step 1.

This procedure ensures that the accepted samples are distributed according to $p(x)$.

1.3. Extension with Proposal Distributions

A more general version of the acceptance-rejection method introduces a proposal distribution $g(x)$ and a constant M such that

$$p(x) \leq Mg(x), \quad \forall x. \quad (5)$$

The algorithm is then modified as follows:

1. Sample

$$x \sim g(x). \quad (6)$$

2. Sample

$$u \sim \mathcal{U}(0, 1). \quad (7)$$

3. Accept x if

$$u \leq \frac{p(x)}{Mg(x)}, \quad (8)$$

otherwise reject and repeat.

This formulation is more efficient when $g(x)$ approximates the shape of $p(x)$, reducing the number of rejected samples.

1.4. Physical Interpretation

From a physical perspective, the acceptance-rejection method allows us to simulate random variables distributed according to a given differential rate. In the context of particle physics, if the differential decay rate $\frac{d\Gamma}{dx}$ is known, it can be interpreted as a probability density (up to normalization), and the method enables the generation of synthetic decay events that reproduce the expected physical distribution.

References

- Czarnecki, A., Dowling, M., Garcia i Tormo, X., Marciano, W. J., & Szafron, R. (2014). Michel decay spectrum for a muon bound to a nucleus. *Physical Review D*, 90(9). <https://doi.org/10.1103/physrevd.90.093002>
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